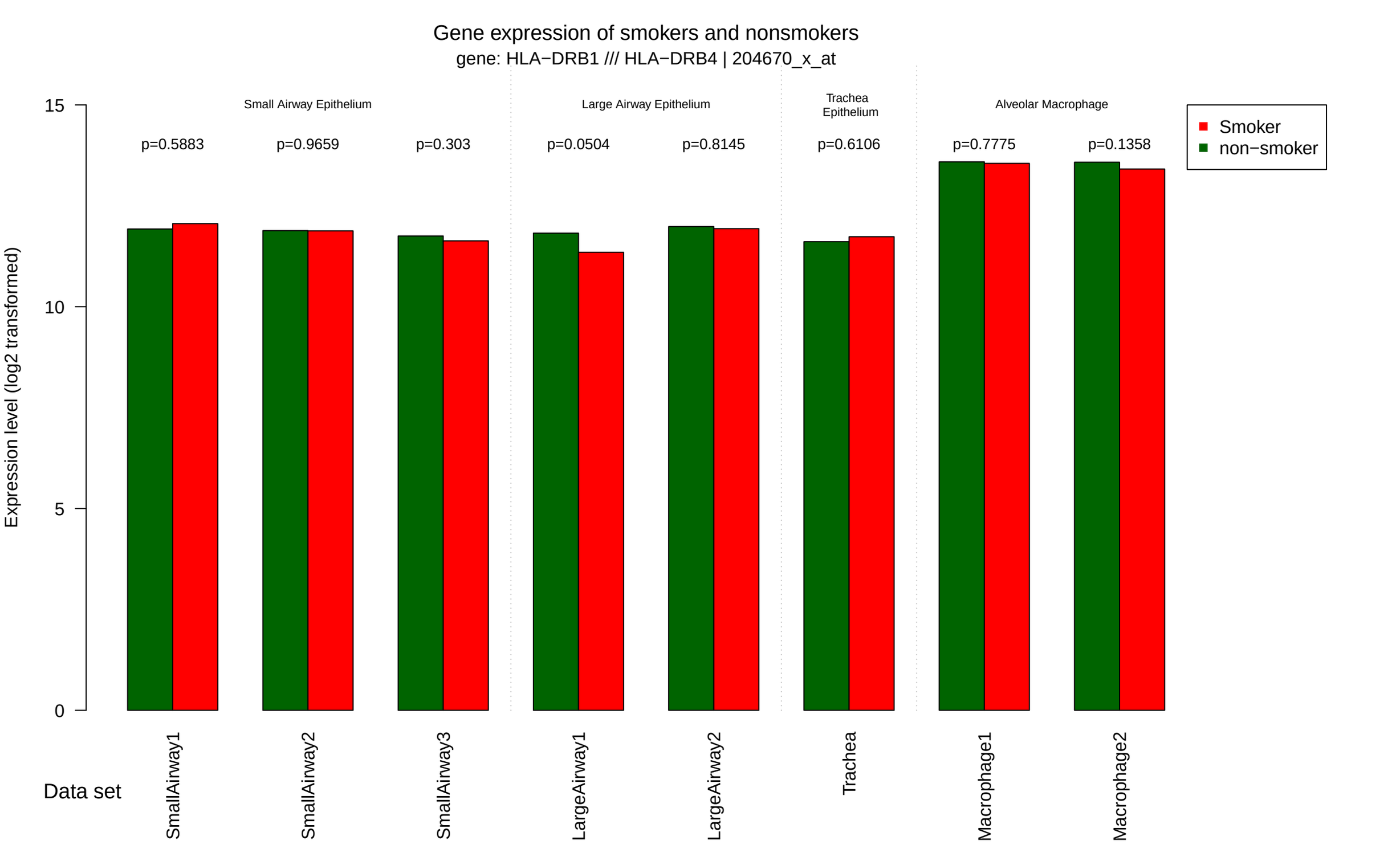
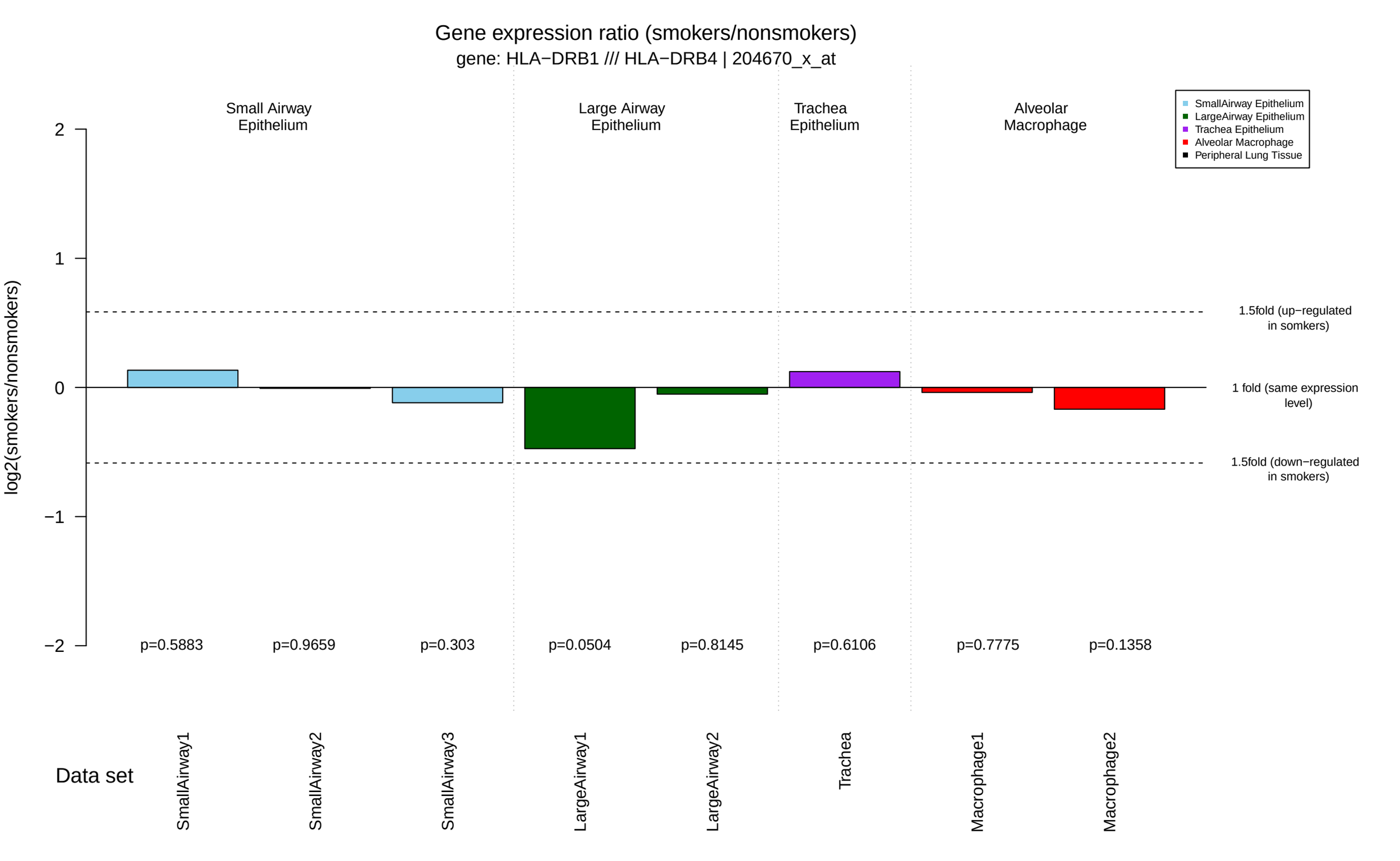
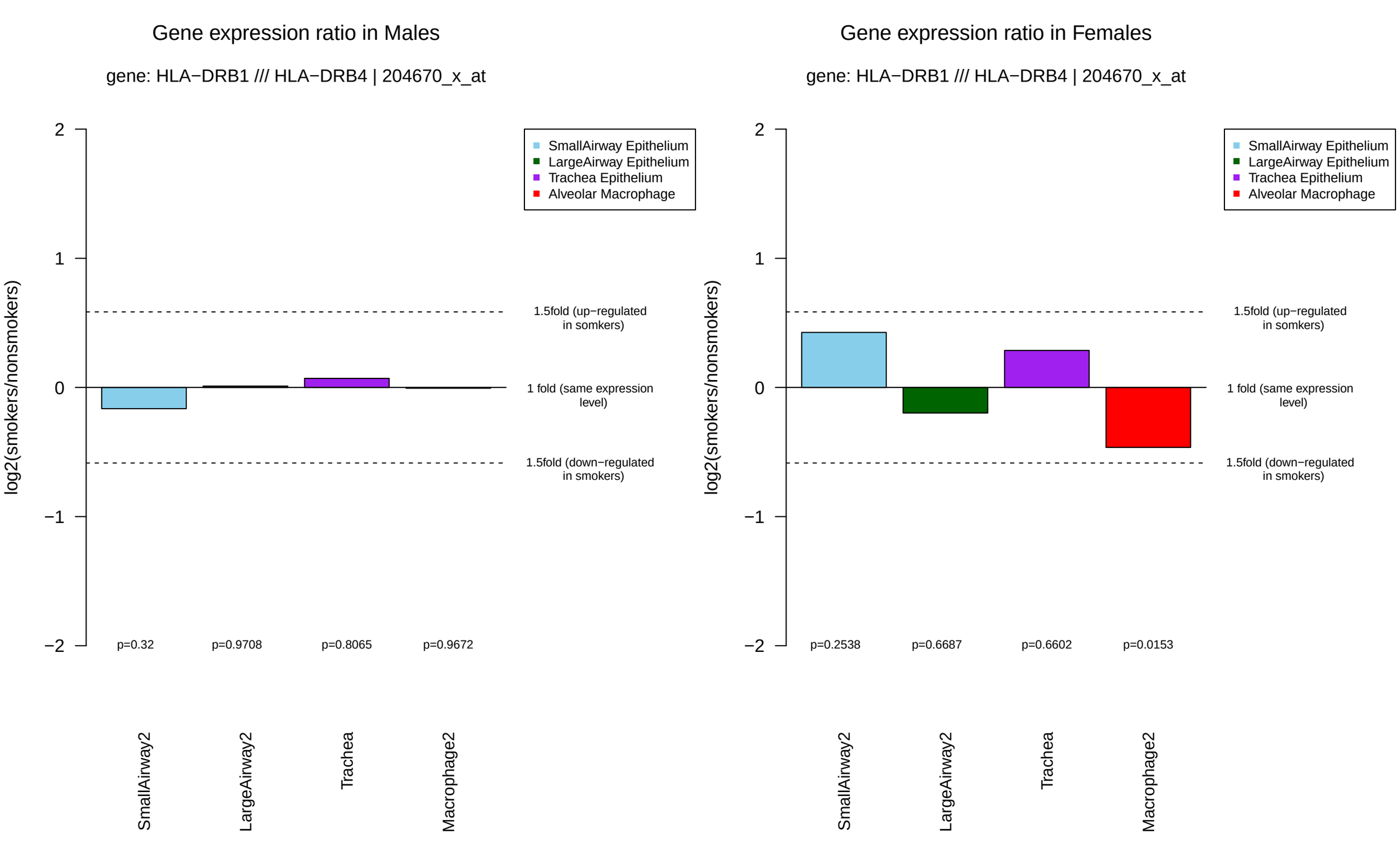
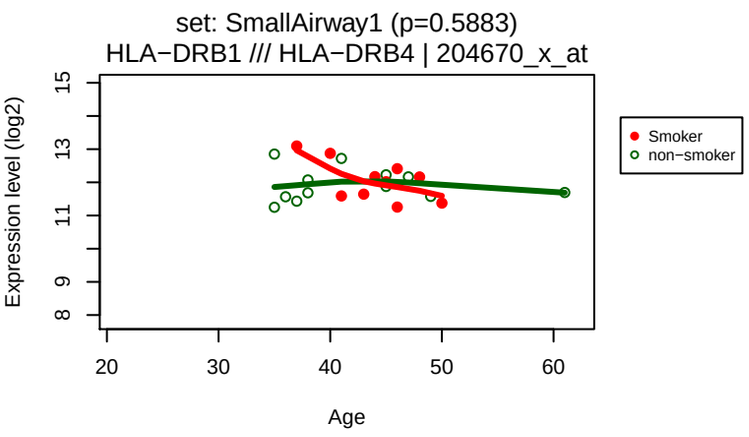
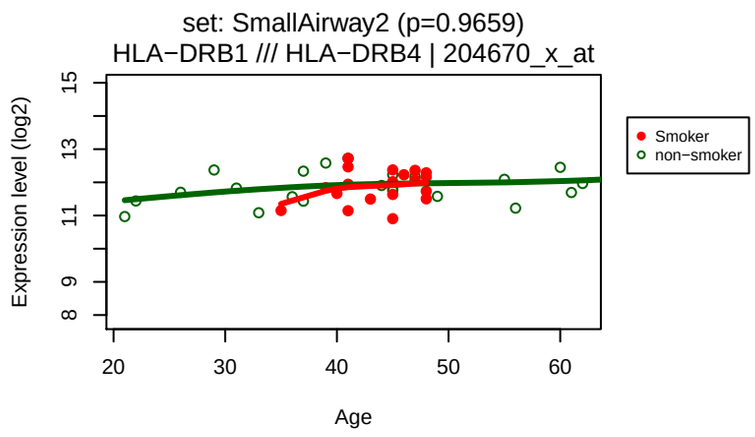
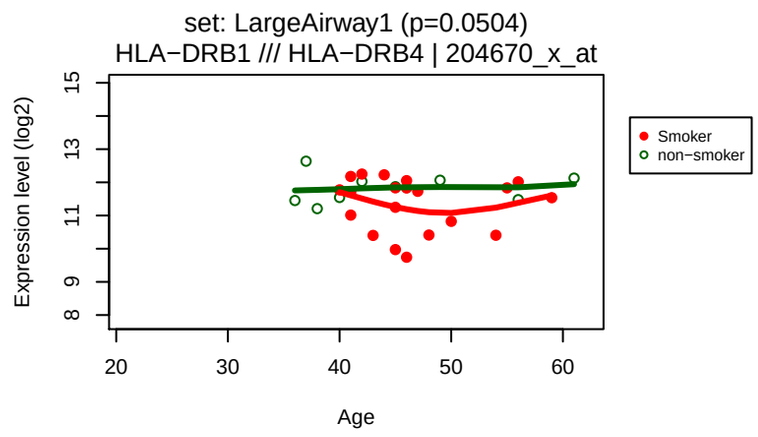

Gene expression of smokers and nonsmokers
gene: HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5883
p=0.9659
p=0.303
p=0.0504
p=0.8145
p=0.6106
p=0.7775
p=0.1358
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

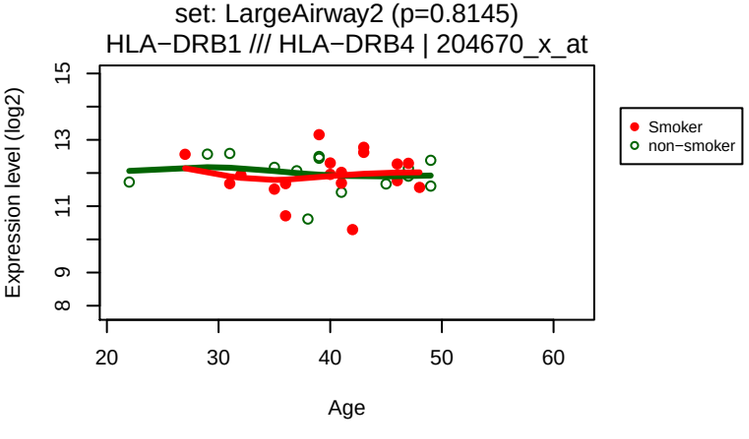

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5883
p=0.9659
p=0.303
p=0.0504
p=0.8145
p=0.6106
p=0.7775
p=0.1358
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

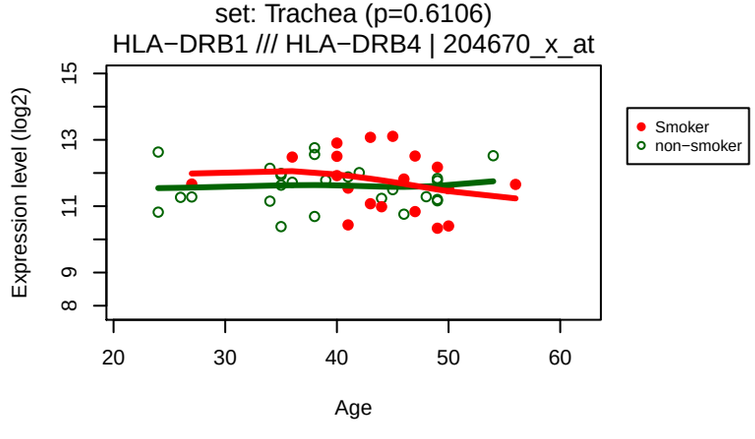

Gene expression ratio in Males
gene: HLA−DRB1 /// HLA−DRB4 | 204670_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.32
p=0.9708
p=0.8065
p=0.9672
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DRB1 /// HLA−DRB4 | 204670_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2538
p=0.6687
p=0.6602
p=0.0153
SmallAirway2
LargeAirway2
Trachea
Macrophage2

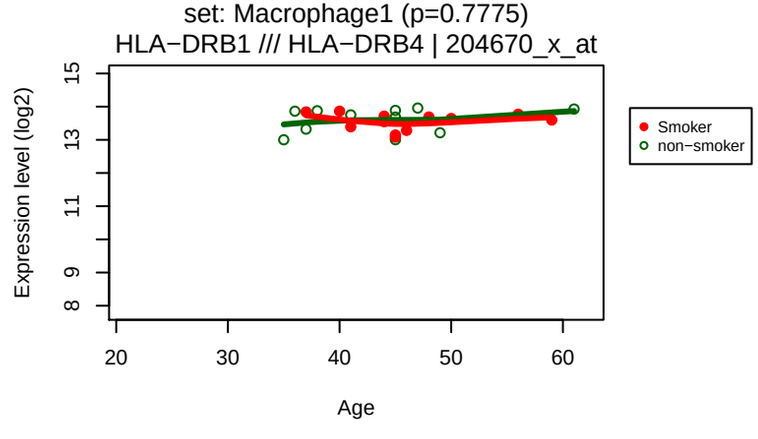

set: SmallAirway1 (p=0.5883)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker

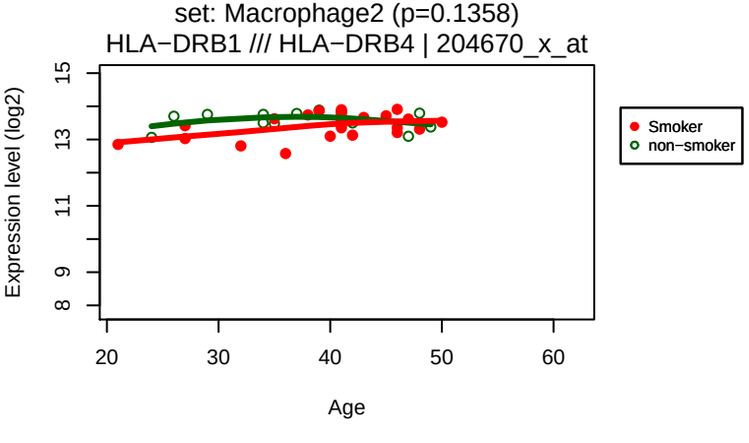

set: SmallAirway2 (p=0.9659)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker

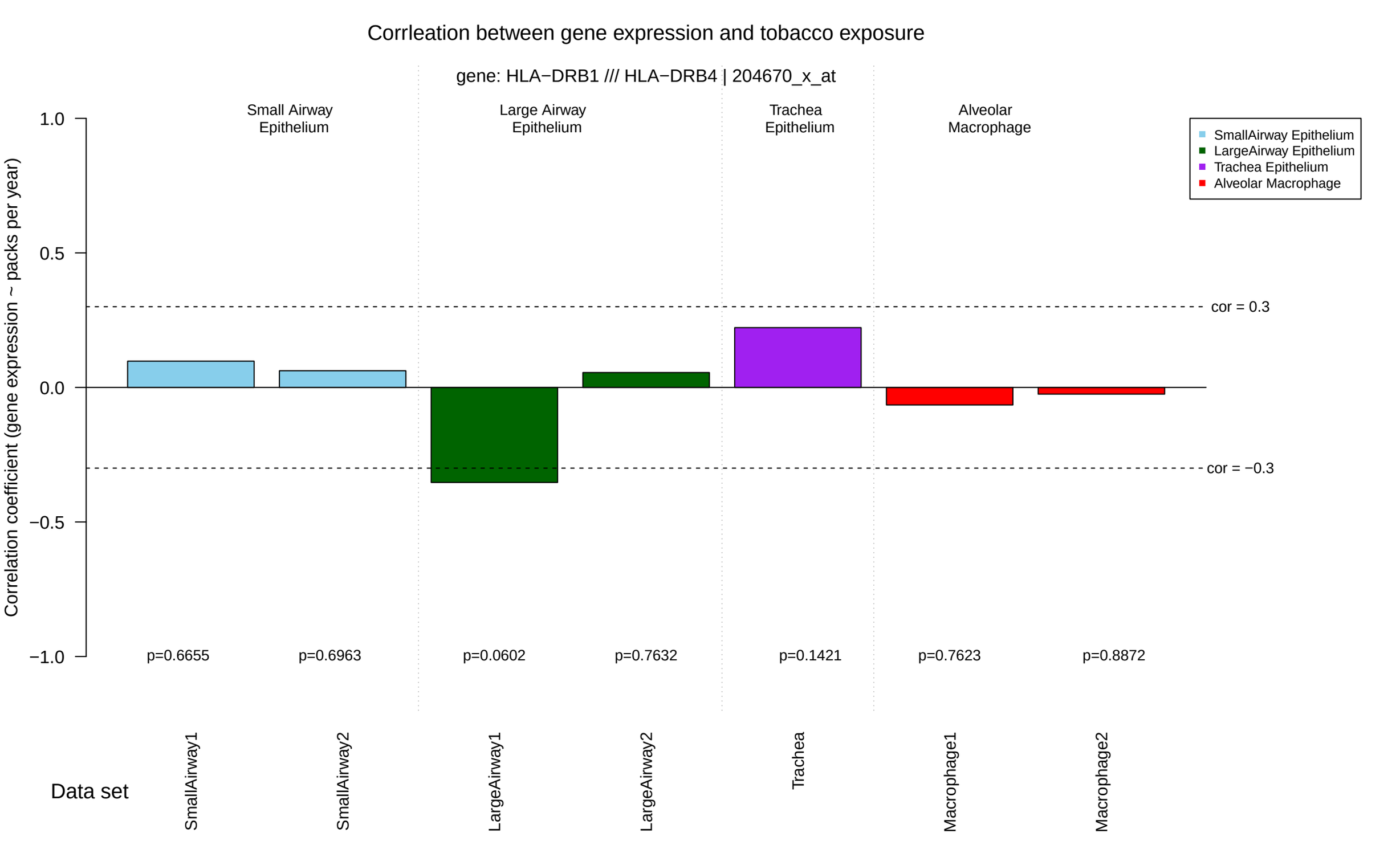

set: LargeAirway1 (p=0.0504)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8145)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6106)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7775)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1358)
HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DRB1 /// HLA−DRB4 | 204670_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6655
p=0.6963
p=0.0602
p=0.7632
p=0.1421
p=0.7623
p=0.8872
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2